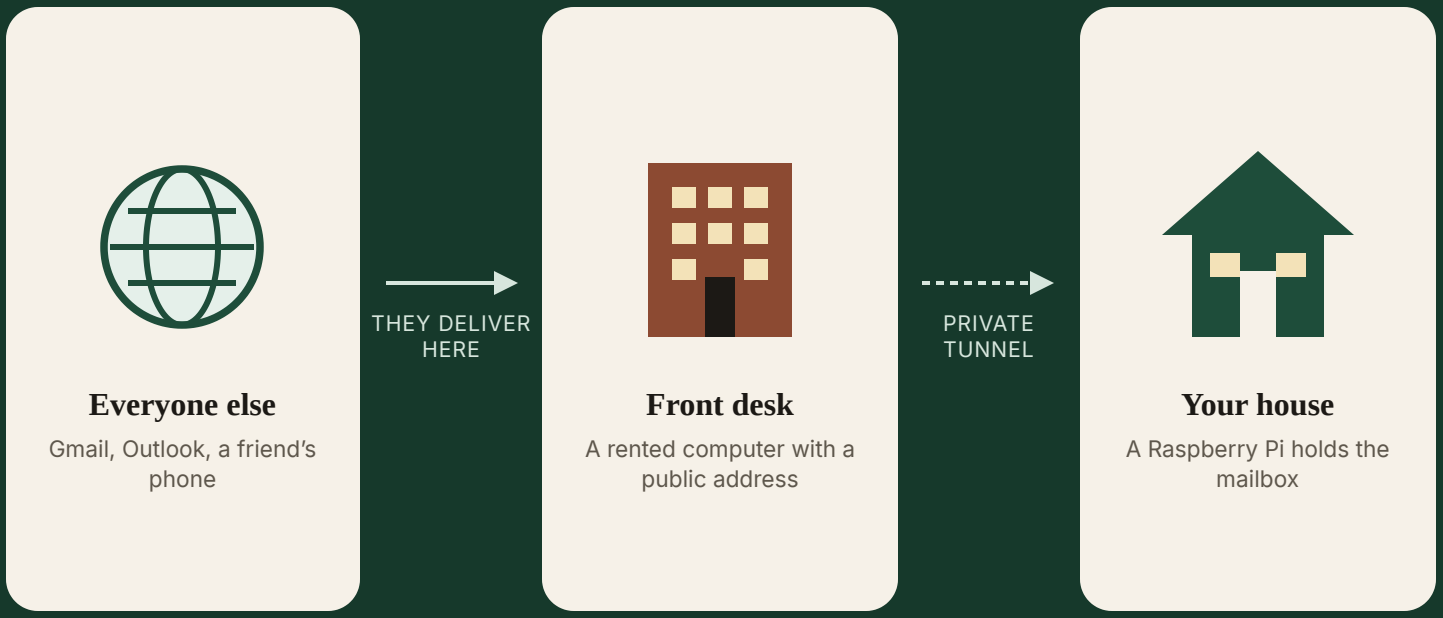
A PICTURE GUIDE

Your mail, at home

What an email server is, explained with pictures, for someone starting from zero. Then the plan for running one on a Raspberry Pi.



1

The letters live on the small computer at home.

2

A rented computer is the door the rest of the world uses.

3

Your phone stays a phone. It reads the mail. It does not store the master copy.

The Pi at home keeps the mail. A rented computer with a steady public address is the door. A private tunnel connects them.

START HERE

Email is a letter

An email server is the room where letters are kept. Your phone is the window you look through. This guide is about the room.

ON PAPER

A letter in a mailbox

1. You write a letter and put a street address on it.
2. The post office looks up that street.
3. The letter waits in a mailbox until someone opens it.

ON A COMPUTER

The same three steps

1. You write in a mail app and put an email address on it.
2. A worldwide address book looks up the name after the @.
3. The letter waits on a mail computer until a phone fetches it.

An address has two halves

THE PERSON

alex

One mailbox, with that person's letters inside.

@

THE DOMAIN

example.com

The name you own. Like the street.

THE BUILDING, WHICH IS A DIFFERENT NAME

mail.example.com

The computer that handles mail for the street. People write to alex@example.com. Computers walk to mail.example.com.

example.com is a made-up name reserved for stories like this one. Yours will be a name you choose and buy.

THE CAST

Four jobs, four different things

These get mixed together. They are four separate roles. The pictures on the later pages use the same shapes.



Alex
You, in this story.



Sam
A friend who uses Gmail.



The letter
One email, traveling.



The window

The phone or laptop where you read and write. It keeps a copy for you to look at. The master letters live somewhere else.

YOUR APPS, UNCHANGED



The room

The server. A computer that stays on and keeps every letter. In this plan, that computer is a Raspberry Pi in your house.

THIS IS THE EMAIL SERVER



The address book

The internet's phone book, called DNS. One line in it says where letters for your domain should be delivered.

THAT LINE IS CALLED MX



The front door

The public address other companies will actually walk up to. It needs to be steady, reachable, and labeled with the right name.

USUALLY A RENTED COMPUTER

> **The whole project, in one line.**
>
> Put the room in your house. Put the front door on a rented computer that the world can find. Connect them with a private tunnel.

THE HARD PART

A house is a fine room and a poor front door

The Pi can keep letters. That part is ordinary. Other companies decide whether to hand you a letter, or accept one from you, based on the door.



Your house
Pi, mailbox, letters

LOCKED



The street
Home internet sits between

UNTRUSTED



Gmail
Will it accept the letter?

1

The mail door is often locked

Home internet companies often block the door computers use for mail. If that door is shut, nobody can deliver, and you cannot send.

2

The address moves, or is shared

A home address can change. Sometimes many houses share one address, so there is no door of your own on the street.

3

The name on the door does not match

Receivers ask, “Who lives at this address?” They want the answer to be mail.example.com. Home providers almost never publish that.

Four checks, before any software

1	**Is the address shared?**	Compare the address your router shows with the address a “what is my IP” site shows. If they differ, the world cannot knock on your house.
2	**Can outsiders reach the mail door?**	From a phone on cellular data, outside your wifi, see whether your house answers on the mail door.
3	**Can the Pi send outward?**	From the Pi, see whether it can open the mail door of a big mail company. Many home lines block this too.
4	**Will they print your name?**	Ask the internet company to publish your address as mail.example.com. This is called reverse DNS. Home companies rarely agree.

THE PLAN

A front desk downtown, a mailbox at home

Rent a tiny computer in a data center. It has a public address that stays put, and its company will print the matching name. Your Pi keeps the letters and opens a private tunnel out to that desk.



Sam, and every other mail company

They only need one public place to walk up to.

They look up example.com and deliver to the front desk

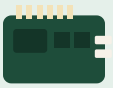


Front desk

A small rented computer. Steady address. Name on the door matches.

`mail.example.com`

Private tunnel. The Pi opens it from inside the house, so visitors never have to find you.



Raspberry Pi, at home

The only mailbox. Every letter is filed here, on a real disk.

When you open your mail app, the phone fetches a copy



Your phone

Same apps you use now. They talk to the front desk, which passes them through to the Pi.

If the Pi is off, the letters wait at the desk and come home when it is back.

The desk passes mail through. It is not a second mailbox. A few homes can skip the desk, if the four checks on the previous page all pass. Most homes use the desk.

FOLLOW ONE LETTER

Sam writes to Alex

Sam’s letter is addressed to alex@example.com. Watch where it walks.

1



Sam writes the letter

On a phone, in Gmail. The address is alex@example.com. Sam taps send.

2



Gmail looks up the street

The address book says mail for example.com goes to mail.example.com, the front desk. That line in the book is called an MX record.

3



The letter arrives at the front desk

Gmail knocks on the public mail door. The desk accepts the letter because it is addressed to someone at example.com.

4



It crosses the private tunnel

The desk walks the letter through the hallway to the Pi. The hallway is called WireGuard. Your house can open it from the inside.

5



The Pi files it

A junk-mail sorter scores it. The letter is stored on the disk at home. Alex’s phone picks up a copy the next time the app is opened.

> **What Alex notices.**
>
> A new message, in the same kind of app as always. The tunnel, the desk, and the Pi stay out of sight.

FOLLOW ONE LETTER

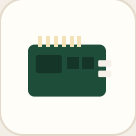
Alex writes back

Sending is the direction that gets people into trouble. Gmail trusts a letter when the address it came from matches the name on the door.

1 

Alex taps send

The phone connects to a sending door and gives a password. Strangers cannot use this door. The password proves it is Alex.

2 

The Pi takes the letter

A copy is filed at home, in Sent. The Pi prepares the letter for the outside world.

3 

The Pi stamps a seal

The seal is called DKIM. Later, Gmail can check the seal and see the letter really comes from example.com.

4 

The front desk carries it out

The Pi hands the letter to the desk. The desk delivers it from the public address, the one whose name matches mail.example.com.

5 

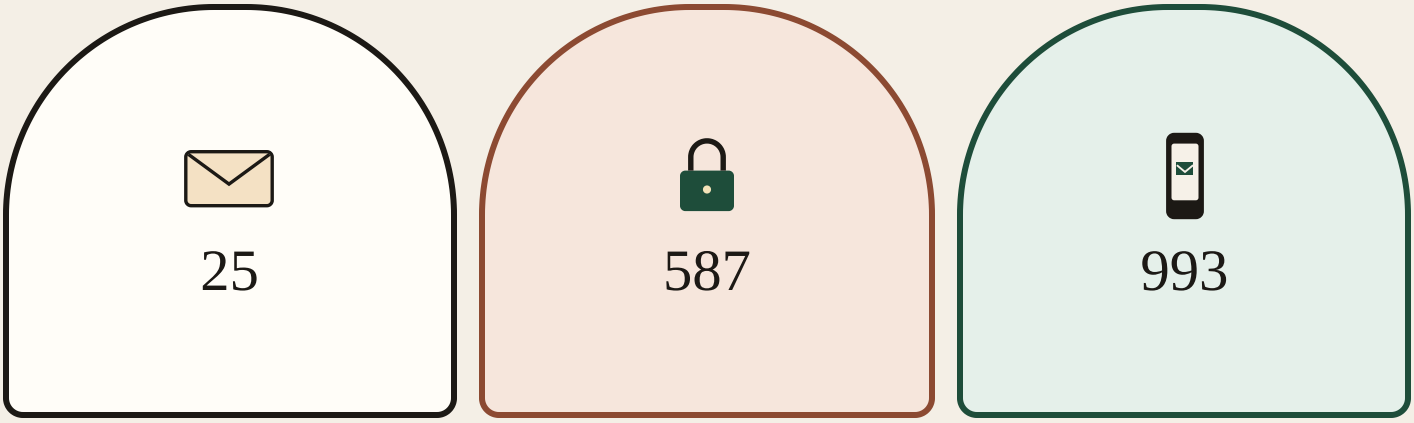
Gmail checks the guest list

The guest list is called SPF. It says the front desk is allowed to send for example.com. The house address is not on that list, on purpose.

> **A brand-new door is a stranger.**
>
> The first letters to Gmail sometimes land in spam. Ordinary mail, sent and answered like a person, is how that door earns a reputation. Bulk mail is a different project and does not belong here.

A computer has numbered doors

A port is a numbered door. Mail uses three. Each one has a different visitor.



Public mail slot

Other companies deliver here. The door accepts letters for people at your domain, and turns away mail that asks to be passed on to someone else.

Your sending window

Alex's phone uses this door to hand over a new letter. It asks for a username and a password. The trip is locked so nobody on the way can read it.

The reading room

The phone uses this door to fetch letters. It also asks for a password, and the trip is locked. This way of reading is called IMAP.

> **In this plan, all three doors are on the front desk.**
>
> The desk passes each visitor through the tunnel to the Pi. The Pi is the one who files the letter, checks the password, and keeps the mailbox. The lock on the trip is a certificate: a document that says "this really is mail.example.com."

WORDS YOU WILL SEE

The same ideas, with their real names

You already met these in the story. This page is the translation, for when a settings screen uses the short name.



Server

A computer that stays on and waits for requests. Here, the room where mail is kept.



Raspberry Pi

A small, cheap, quiet computer. Ours lives at home and holds the mailbox.



Docker

A box around a program, with everything that program needs. The mail program runs in one box.



VPS, the front desk

A small computer you rent in a data center. “VPS” is the rental name. We call it the front desk.



DNS and MX

DNS is the address book. MX is the line that says where mail for your domain goes.



WireGuard

The tool that builds the private tunnel between the Pi and the front desk.



SPF

The public guest list. It names who may send mail for your domain. Ours names the front desk.



DKIM

The seal the Pi stamps on outgoing letters. Receivers check the seal against a public key in the address book.



DMARC

A public note that says what to do when the guest list or the seal fails. We start by asking for reports only.



TLS certificate

The lock. It proves the server is mail.example.com and keeps the letter private on the wire.



Reverse DNS

The question “who lives at this address?” answered with mail.example.com. The front desk’s company sets this.



Mailbox

One person’s pile of letters on the disk. alex@example.com is one mailbox.

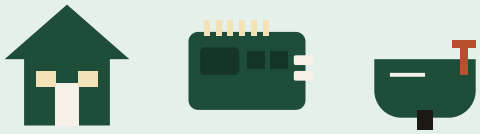
What sits where

Two computers, two jobs. Mail is stored in one place.

AT HOME · GREEN

The Raspberry Pi

- Keeps every mailbox.
- Checks Alex’s password.
- Stamps the seal on outgoing letters.
- Sorts junk mail.
- Opens the tunnel out to the desk.



DOWNTOWN · BRICK

The front desk

- Owns the public name and address.
- Has the three doors: 25, 587, and 993.
- Passes visitors through the tunnel.
- Delivers Alex’s letters to Gmail and the rest.
- Holds new letters only while the Pi is off.



Inside the Pi, one Docker box with four helpers

`Postfix`
Receptionist
Takes letters in and hands letters out.

`Dovecot`
Filing clerk
Lets the phone read the mailbox.

`Rspamd`
Junk sorter
Scores spam and stamps the seal.

`Fail2ban`
The guard
Slows people who guess passwords.



The little card

Comes with the Pi and runs the system. Letters are written all day, and that wears a little card out. The card keeps the operating system.



A USB disk

A real SSD, plugged into the Pi. The mailboxes, the Docker box, and the seal’s private key live here. 128 GB is plenty for a household.

THE ORDER

Build it in this order

Each step has something you can see. Stay on a step until that thing is true. The mail program comes after the door works.

1. **Learn what the home line allows**
   Do the four checks. They decide whether you need the front desk. On a typical home line, you do.

2. **Choose a domain, and a desk**
   Buy a name whose address book you can edit. Rent the front desk only from a company that will open the mail door and print the matching name.

3. **Give the Pi a real disk**
   Plug in the USB SSD. Restart the Pi and confirm the disk is still there. Letters will be written to that disk.

4. **Start the mail box, in private**
   Run the Docker program on the Pi. Make two test mailboxes. Send a letter from one to the other, inside the house, before the world is involved.

5. **Publish the address book and the lock**
   Add the MX line, the guest list, the seal's public key, and the DMARC note. Get the certificate for mail.example.com.

6. **Send one letter to a real Gmail account**
   Read the details Gmail shows. The sending address should be the front desk. Then reply, and watch that reply land on the Pi.

7. **Put it on your phone**
   Add the account in the normal mail app. Reading uses door 993. Sending uses door 587. Both ask for the password.

8. **Make a backup, and open the backup**
   Copy the disk's mail, the account list, and the seal's private key somewhere else. Restore that copy once, and open a letter from it, before you depend on it.

BEFORE WE BUILD

Four decisions, then we can start

The pictures stay the same. These answers change the details.

1. **What domain do you want?**
   You need a DNS host that lets you edit MX and TXT records. SPF, DKIM, and DMARC are all TXT records.

2. **Is a small rented computer okay?**
   It is often a few dollars a month. On a normal home connection, this is the decision that lets mail arrive and be trusted.

3. **Who gets a mailbox?**
   One person is the simple start. A household means a password and a pile of letters for each person, plus a backup that matters.

4. **Is the Pi free for this job?**
   A Pi 4 or 5 with 4 GB of memory is enough when mail is the main thing it does. If it already runs other things, 8 GB is the comfortable size.

After the mailbox works

- Reading mail in a web browser
- A second domain
- Filters and folders
- Stricter DMARC, once reports look clean

A catch-all address accepts mail for any name at your domain, and it fills with junk. Mailing lists are a different system. Both wait.

The sentence to keep

The Pi at home keeps the mail. A rented computer with a steady public address is the door. A private tunnel connects them. Your phone works the way it does today.